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Analysing the synergies between Multi-agent Systems and Digital Twins: A systematic literature review

Elena Pretel, Alejandro Moya, Elena Navarro *, Víctor López-Jaquero, Pascual González

LoUISE Research Group, Computing Systems Department, University of Castilla- La Mancha Avda. España s/n, 02071, Albacete, Spain

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ABSTRACT

Context: Digital Twins (DTs) are used to augment physical entities by exploiting assorted computational approaches applied to the virtual twin counterpart. A DT is generally described as a physical entity, its virtual counterpart, and the data connections between them. Multi-Agent Systems (MAS) paradigm is alike DTs in many ways. Agents of MAS are entities operating and interacting in a specific environment, while exploring and collecting data to solve some tasks.
Objective: This paper presents the results of a systematic literature review (SLR) focused on the analysis of current proposals exploiting the synergies of DTs and MAS. This research aims to synthesize studies that focus on the use of MAS to support DTs development and MAS that exploit DTs, paving the way for future research.
Method: A SLR methodology was used to conduct a detailed study analysis of 64 primary studies out of a total of 220 studies that were initially identified. This SLR analyses three research questions related to the synergies between MAS and DT.
Results: The most relevant findings of this SLR and their implications for further research are the following: i) most of the analyzed proposals design digital shadows rather than DT; ii) they do not fully support the properties expected from a DT; iii) most of the MAS properties have not fully exploited for the development of DT; iv) ontologies are frequently used for specifying semantic models of the physical twin.
Conclusions: Based on the results of this SLR, our conclusions for the community are presented in a research agenda that highlights the need of innovative theoretical proposals and design frameworks that guide the development of DT. They should be defined exploiting the properties of MAS to unleash the full potential of DT. Finally, ontologies for machine learning models should be designed for its use in DT.

1. Introduction

In the latest decades, the evolution of technologies such as the Internet of Things (IoT), Artificial Intelligence (AI), Virtualization or Cloud Computing are advancing at high speed. The interest in Digital Twins (DTs) has grown considerably in recent years, in both academia and industry, due to its significant potential in many application fields, such as industrial production, manufacturing, health, smart cities and transport, aerospace, energy, business world, etc. [1]. The growth in the number of publications related to DT has become exponential in the last years. As a result of the continuous development of science and technology, DT is a constantly evolving concept. Currently, there are still many challenges to be solved in promoting DT to practical applications. One of the fundamental challenges in DT-related research is its design, modelling, and representation. The main difficulty lies in developing a model that represents with high fidelity all the relevant attributes of the physical assets.

Among the various paradigms that can be used to address this challenge, MAS stands out for its many similarities with DT [2]. MAS are agent-based systems which act as an external entity operating in a particular environment while exploring and collecting data [3]. MAS, with its ability to model autonomous and cooperative behaviours between agents, offers a flexible and dynamic structure that can adapt to changes and needs in real time. This approach may not only improve the accuracy and fidelity of DT, but also enable new forms of interaction between digital and physical components. The potential of integrating DT and MAS has already been explored in the literature and the synergy between these two approaches has potential to be applied to different domains, from smart manufacturing [4] to smart city planning [5] and e-health [6]. For example, in [4] authors describe a DT that whenever it

* Corresponding author.
E-mail address: elena.navarro@uclm.es (E. Navarro).

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perceives an anomaly in its behaviour, it can stop production automatically to avoid further losses. This self-managed DT is a reactive agent that perceives and acts on its environment by responding in a timely manner to the changes occurring around it. Here it is possible to see how the DT can take advantage of agent properties to provide its own properties. These synergies will be analysed during this SLR.

In this paper, we studied the synergies of MAS and DTs. Several authors have studied the idea of using DT and MAS together [7,8]. We differ from these studies mainly in that this paper focuses on analysing how MAS can support the development of DT rather than just how DT and MAS can be combined. In other words, we focus on analysing the synergies between both technologies to see how DT can be developed using MAS. No previous work has been carried out that focuses on investigating the synergies between DT and MAS from this point of view: MAS to support the development of DT. This paper presents a systematic literature review of existing proposals using MAS to develop DT along with MAS that use DTs. Therefore, this paper aims at identifying and synthesizing current studies that focus on MAS and DTs and to identify the development technologies and the way in which knowledge is represented, as well as the corresponding challenges and next steps to be taken in this field. Therefore, the contributions of this paper are as follows:

- A first systematic and comprehensive study of DTs developed with MAS for different domains, the development technologies applied and their purposes.
- A synthesis of observations on the research landscape of DTs in conjunction with MAS that can guide future research.
- A novel guide to developing the properties of DT by using the properties of MAS guiding researchers and practitioners in decision making on DT design and development.

The rest of the paper is organized as follows. Section 2 describes DT technology and MAS. Next, Section 3 presents the related work. Then, Section 4 describes the methodology used for the study and Section 5 shows the results and the analysis carried out to answer the stated research questions. Section 6 lists which trends have been identified in the recent literature reviewed and discusses these findings with regard to the research questions. Next, Section 6.4 lists the threats to validity this work. Finally, Section 8 presents both the conclusions drawn and our future work.

2. Background

This section describes the two key concepts, DT and MAS, which this SLR is based on.

2.1. Digital Twin

Although DT concept has gained great popularity in recent years, the concept is not completely new, since it was originally coined by Grieves and Vicker in 2003 [9]. Their proposal, as illustrated in Fig 1, was made up of three elements: i) a real space, ii) a virtual space, and iii) a link between real and virtual space for the bi-directional flow of data between both spaces for the convergence of the real and virtual world. This model proposes a preliminary idea for dealing with the real and virtual world in DT, but it does not prescribe neither how they interact nor indicate which technologies should be used to achieve it.

As discussed in [7], there is a wide variety of definitions of DT in the literature. Several attempts have been made aiming to provide a widely accepted definition. For example, in [2,7,11,12] have collected different definitions of DT drawn from other proposals, as well as the evolution of DT concept. This paper adopts the definition of DT provided by Minerva et al. [2]. These authors agree, at a high level, about defining a DT as a synchronized digital representation (logical object, LO) of an element, process, or system of the physical space (physical object, PO). A LO interacts continuously with its PO, sharing data in real time, synchronizing the physical and cyber worlds so when something happens, the LO is updated to reflect the new information. In short, DT can be defined as a virtual representation (LO) of an object, process or system (PO) that spans its life cycle, is updated with real-time data, and uses simulation, machine learning and reasoning to aid in decision making. Minerva et al. [2] also identify the main properties that a system must satisfy to be considered as a DT, that is, those that characterize DT.

The concept of DT has evolved since its inception. Originally, a DT was focused solely on modelling physical objects in the virtual world. Today, a DT usually refers to an approach capable of simulating the dynamics, monitoring the state and controlling or predicting the behaviour of the physical object. DTs are being used in a wide variety of areas such as manufacturing [13], healthcare [14], agriculture [15], etc.

DT can be classified into different types based on the interaction between the PO and LO [12,16]. Thus, there are three subcategories (Fig 1):

- Digital Model: data between the PO and LO are exchanged manually, so changes in the state of the PO are not reflected in the digital one directly, and vice versa.
- Digital Shadow: data from the PO flows to the digital one automatically, but the reverse is manual. Thus, any change in the PO can be seen in its LO, but not vice versa.
- DT: there is an automatic bidirectional flow of data between the PO and LO. Therefore, changes in either the LO or the PO directly cause changes in the other.

This definition is based on an exhaustive review of the main definitions, properties and implementations of the DT concept and study its applicability to IoT application scenarios, emphasizing the context of use in which DT operates. This last point is important, since it is almost impossible to fully represent a PO with all its attributes, and also, this may be unnecessary as not all these attributes might be relevant in the context of use of a specific DT. The context of use of the DT corresponds to the particular conditions in which it operates. According to Minerva et al. [2], a DT consists of a *physical object* (PO) and one or more complete software representations called *logical objects* (LOs). This DT definition is consistent with the three elements proposed by Grieves: i) physical entity, ii) virtual entity and iii) relationship between them. Moreover, as part of the definition of DT, in [2] identified also which properties a proposal should have in order to be considered a DT. These DT properties are classified into two groups as follows.

The first set of properties, called *essential properties*, ensures that the

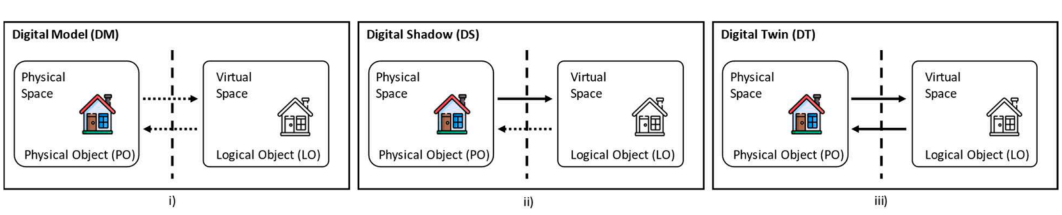

Fig 1. (i) Digital Model; (ii) Digital Shadow; (iii) Digital Twin (adapted from [10]).

DT represents the PO properly and acts in a similar way to it in a specific context. This set is made up by three properties:

- *DT1.*Representativeness *and contextualization*: fully representing the PO as a virtual object can become a cumbersome task, and sometimes even unnecessary. Therefore, the LO should have only those attributes of the PO considered relevant according to its context of use. For example, if an application uses a DT to monitor a domestic radiator, the only valuable information is that related to its temperature and internal pressure. Attributes like colour or brand may have little or no relevance in the context of a home application.
- *DT2.Reflection*: this property determines that every significant attribute of the PO must be represented properly in the LO. The LO updates its state after collecting data from the physical world, being this state complete when all its data have been updated. Thus, the state of the LO is the same as that of the PO. Considering the IoT application domain, this property refers to measurable aspects of the PO. These measures are then represented by a set of values that can clearly reflect the specific status of the PO in the analysed context. For example, the attribute temperature of the domestic radiator DT can be represented as {*timestamp, location, value*}, this could be set as {2024-04-11T10:00:00Z, [40.7128, -74.0060], 21°C}, indicating that at 10:00 AM on 11 April 2024, at the specified location (latitude 40.7128, longitude -74.0060), the temperature was 21°C.
- *DT3.Entanglement*: this property refers to the communication between the LO and PO. The exchange of information between them must be instantaneous, in real time. There are two types of entanglement: (a) weak entanglement: communication is unidirectional from the PO to the LO. For example, the temperature sensor of the radiator sends data to the LO; and (b) strong entanglement: communication is bidirectional and the LO has the ability to modify or update the state of the PO. Thus, the temperature sensor of the radiator sends data to the LO and this sends commands to an actuator which automatically switches the domestic radiator (PO).

The second group of properties gathers those that provide added value to the DT. All these properties that add value to the DT can be included in the defined DTs as needed in each context of use. These properties that add value are defined as follows:

- *DT4.Replication*: this property describes the ability to replicate a PO into different environments. A PO can be represented in different digital worlds according to different contexts of use, that is, different replicas (LOs) can be created in different environments and may have different characteristics to make it more suitable for a specific context. Let us continue with the example of the domestic radiator DT, many LOs (replicas) of the domestic PO radiator can be created. Each replica can be adapted to the specific needs of different environments or applications. For example, one replica may focus on energy optimisation, analysing historical and real-time data to minimize energy consumption. Another replica can specialize in user comfort, adjusting the temperature according to recorded personal preferences.
- *DT5.Persistency*: this property indicates that the DT must be always available, that is, it must be resilient and persistent over time. In the case of malfunctioning or other problems with the PO, the LO should be the source of information for re-establishing and synchronizing the PO to an acceptable and meaningful state. Continuing with the example, in order to ensure the persistence of the LO, multiple replicas of this LO can be deployed in different server or cloud environments, ensuring that if one environment fails, another can take over without service interruption and is always up and running when the PO needs it.
- *DT6.Memorization*: this property describes "the ability to store and represent all the present and past data relevant for the DT". All these data reflect the behaviour of the PO and the context in which it acts. The use of these data is twofold: to understand the behaviour of the PO within its environment, and to predict its possible behaviour in such environment as well as in others. Again, in the context of a DT to control a domestic radiator, where the relevant attributes are temperature and internal pressure, the entire data history of these two attributes must be stored to facilitate their analysis and exploitation later on.
- *DT7.Composability*: generally, in real life, POs are often an aggregation of different entities. This property refers to the ability to group several LOs into a composite one and then observe and control the behaviour of the composite LO as well as the individual LOs. When the DT represents a complex entity or system, the integration of new components must be done in a simple way. For example, the lightning system and the heating system of a smart building must be integrated and managed separately as well as a whole system.
- *DT8.Accountability/Manageability*: this property describes the ability to manage DT completely and accurately. For example, in case of a failure in the PO, the LO must facilitate its recovery by displaying all the most recent important functional values. Moreover, the LO must have functionalities for its self-management and recovery. In case of physical radiator failure, the DT acts as the main source of information for diagnosis and resolution of problems. For example, if a reduction of the heating efficiency is detected, the DT can analyse historical data to identify failure patterns and suggest solutions based on those learnings.
- *DT9.Augmentation*: this property indicates that new attributes and functionalities can be added to the LO as required. That is, if the context of the DT changes, extra attributes may be relevant to be included in it. Thanks to this property, by using data and new technologies such as Big Data, it is possible to enhance the PO by adding new attributes and functions to it. For example, the LO of the domestic radiator can be provided with data processing capabilities to provide different profiles of behaviour according to the agenda of its users.
- *DT10.Ownership*: this property defines two types of owners. The first one represents who owns the data, as the DT produces a large quantity of data it is important to regulate its use and to know who owns it. The second type of owner refers to who owns the DT and, in particular, the LO. Normally, the PO has an owner, and its virtualization can create a set of LOs that reference the physical one, but they do not necessarily share the same ownership. For example, the data generated by the domestic radiator may belong to the owner of the physical radiator and the DT to the company that manufactured the PO and LO and manages them.
- *DT11.Servitization*: this property refers to the ability to provide the physical world with new services and functionalities created in the LO. In this way, the DT is now seen by customers as an attractive service in terms of its functionalities and not just as an object. For example, in a DT in the field of manufacturing, this property allows the product to be personalized, making it much more attractive to the user. This personalization is one of the main values added to the physical product as it is now adapted to the user's needs rather than the user having to adapt to the product.
- *DT12.Predictability:* given the large amount of data sets that the LO has at its disposal it can interact with other LOs and POs. This property refers to the ability of the LO to simulate its behaviour and interactions with other objects during a specific period in a specific environment. This is a growing trend in AI that can be applied to the DT concept. Controlling and simulating the behaviour of complex systems such as cities, factories or logistics networks are examples of how this property of DT is used in industry.

Nowadays, there are still many challenges to be solved to bring DT into practical applications. One of the fundamental challenges in DT-related research is the need to provide support for its development. To solve this problem, in this paper we study whether the MAS may be used to address the DT properties.

2.2. Multi-Agent Systems

Multi-Agent Systems (MAS) are agent-based systems, which operate and interact in a specific environment while exploring and collecting data to solve a complicated task [3]. These agents are commonly used to simulate complex environments in which they have to coordinate and cooperate or compete to achieve common or global goals. In the current literature, there are multiple definitions of agents derived from various specific properties of the agents. After a review of the literature, Dorri et al. [3] propose a generalized definition of agent and MAS considering their fundamental abilities and properties.

Dorri et al. [3] define an agent as an entity that is situated in an environment and detects different parameters that are used to decide what to do according to the agent's goal. The agent performs the necessary actions on the environment based on this decision. The objective of each agent is to solve its assigned task. To achieve this goal, the agent first senses parameters from the environment. With this data, the agent can gather knowledge of the environment, and together with the history of previous actions performed, it decides the appropriate actions to accomplish its assigned task or goal [17]. The following properties enable the agents to solve complex tasks [3]:

- *Agent1.Autonomy*: each agent can independently execute the decision-making process and take appropriate actions.
- *Agent2.Sociability:* agents can share their knowledge and request information from other agents to improve performance in achieving their objectives.
- *Agent3.Reactivity*: indicates that an agent can react to changes within its environment and can maintain interactions with it
- *Agent4.Proactivity*: each agent uses its history, sensed parameters and information from other agents to predict possible future actions. These predictions allow agents to take effective actions that meet their objectives.

As indicated in [3], although an agent working alone is capable of taking actions, the real benefit of agents can only be realized when they work in collaboration with other agents. Multiple agents collaborating to solve a complex task are called a Multiple Agent System (MAS). The most important properties of MAS are described below [3]:

- *MAS1.Leadership*: indicates that a MAS may have a lead agent who defines goals and tasks for the rest of the agents to fulfil a global objective. Depending on the presence or absence of this type of agent, the MAS can be categorized as leader or leaderless. In those MAS with no leader, each agent decides independently its actions according to its own objectives, while in those with a leader, this leader determines the goals for the rest of the agents.
- *MAS2.Decision function*: refers to how the input data affects the output of the agent's decision function. In other words, it refers to how the input parameters (changes in the environment) affect the output (agent's decision). Depending on this, a MAS can be linear or non-linear. In a linear MAS, the agent's decision is proportional to changes in the environment.
- *MAS3.Heterogeneity*: refers to whether the agents that make it up have the same architecture, characteristics, and functionalities. That is, if every agent of the MAS has the same architecture, then the system is homogeneous, while if the architecture of the agents within the system differs, then this system is heterogeneous.
- *MAS4.Agreement parameters*: refers to the number of metrics that the agents must agree upon to meet their objective. Depending on the number of metrics, MAS are classified as first, second or higher order. In first order, agents collaborate to agree on a metric. In second order, agents must agree on two metrics, and a higher-order MAS, agreement between agents occurs when there are more than two metrics that converge to a common value.
- *MAS5.Delay consideration*: enunciates that agents may face various obstacles that delay them in performing tasks. For example, there may be delays in communication between agents to exchange data. Depending on whether these delays are relevant or not, MAS can be classified into two groups: with delay or without delay. The former considers possible sources of delay, while the latter assumes that there are no sources of delay.
- *MAS6.Topology*: refers to the position and relations of the agents. MAS topology can be either *static*, where the position and relationships of an agent do not change during the agent's lifetime, or *dynamic*, where the position and relationships of an agent change as the agent moves, leaves, or joins the MAS, or establishes new relationships with other agents.
- *MAS7.Data transmission frequency:* refers to how often information is shared in a MAS. There are two modes: i) agents collect data that share at predefined time intervals, and ii) agents collect data that share whenever specific events happen.
- *MAS8.Mobility*: refers to the dynamism of the agents in a MAS. Agents can be classified as static agents, when they are always located in the same position in the environment, or mobile agents, when they can move in the environment.

After defining what is an agent and a MAS, as well as the main properties of both, it can be observed that exploiting MAS is a good approach to simulate complex environments and to develop software entities that can represent and mimic the behaviour of real-world entities, similarly to what DTs also pursue. Following the definition of DT proposed by Minerva et al. [2], the DT is composed of a PO and one or more LOs and all these objects can be included in the MAS.

3. Related works

Some authors have already suggested this idea of using MAS to support the development of DT. For example, Minerva et al. [2] discuss the correlation that exists between a LO and an agent as both can represent physical entities. In such paper, it is also highlighted that MAS can be used to simulate the behaviour of a specific environment, which is one of the most significant goals of DTs. On the other hand, as described in [21], a DT consists of multiple entities and virtual environments, where each one has an objective to achieve. [30] also highlight the potential of MAS to control and integrate the DTs. Mariani et al. [18] analyse the possible synergies between agents and DTs, as well as between MASs and networks of DTs, to reason at both the individual and system level. Therefore, as agents bear many similarities to DT, many different aspects of MAS should be considered useful contributions to the development of DTs. This fact has led us to conduct this study.

As stated above, there are relevant literature reviews that explore the field of DTs. Some works [2,7,11,19] have already reviewed the literature to discuss the definitions of DT, the main properties they should have or what areas they have been already developed for. Moreover, several literature reviews can also be found in the field of MAS. Some recent reviews in this field are [3,20,21] that discuss what MAS and agents are, their characteristics, in which applications MAS are used or how agents communicate between them.

We differ from these studies in several aspects. Firstly, it is important to remark that this paper focuses on DTs as a concept linked to Multi-Agent Systems, that is, we analyse here how MAS may be used to support the development of DTs and how MAS may exploit DTs. We are fully aware of the influence that other technologies might have on DT, but these are outside the scope of this paper. The second difference is the methodology applied. This work was conducted as a systematic review without specifying any time frame, so that any existing work related to this topic could be identified. Third, the analysed papers do not correspond to any specific domain of application, so that different fields of application have been also considered in this study.

4. Research methodology

To carry out this study, the guidelines proposed by Kitchenham et al. [22] were applied. As shown in Fig. 2, the literature review was conducted in three different phases.

As Fig 2 depicts, during phase 1 we identified the Research Questions to be addressed by this systematic review that are defined as follows:

- **RQ1**: How do current MAS proposals support those properties expected in a DT? This research question pursues identifying which current proposals of DT are developed with MAS, and to study whether these proposals fulfil the DT properties defined by Minerva et al. [2].
- **RQ2**: How are DT and MAS used in the development of applications? This RQ has been defined to find out what properties of agents and MAS may contribute to support the development of each desirable property of a DT to facilitate its design.
- **RQ3**: What approach do current proposals use to manage knowledge? One concern essential to DT is all the past and present data of the PO (*DT6.Memorization*). Thus, this RQ aims to identify which approaches are being used currently to manage knowledge when MAS and DT are combined.

Then, we defined the search query using the PICO method, as suggested by Kitchenham et al. [22]. The PICO method describes the population, intervention, comparison, and outcomes of the research question to define a search query.

- *Population*: refers to the type of technologies about which the review obtains data. In this case, population are the DTs.
- *Intervention*: refers to the tools used to address the specific topic. In this case, the intervention for this review is multi-agent systems,

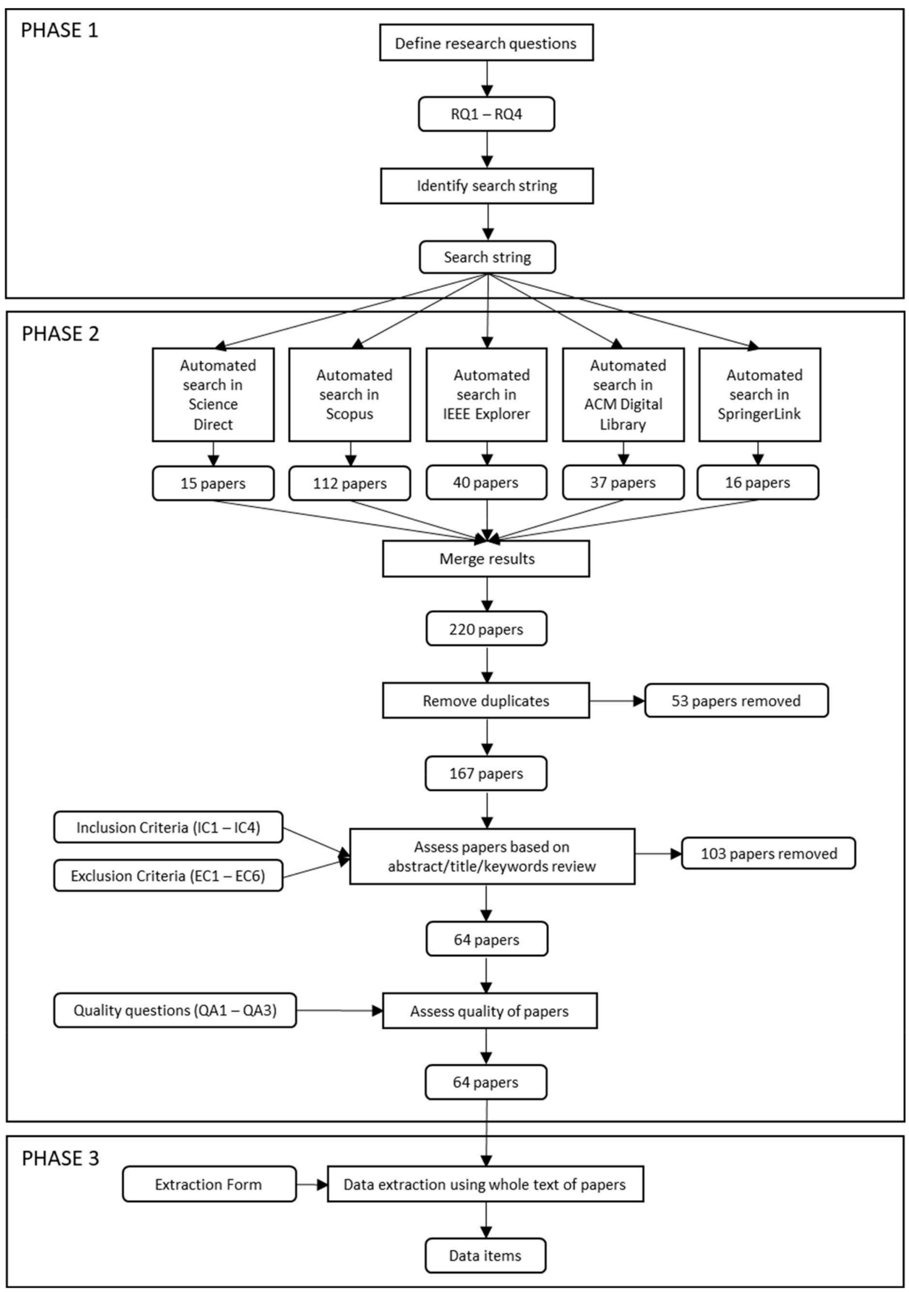

Fig. 2. Search methodology.

since we are investigating how MAS are used to develop DTs and how MAS use DTs.

- *Comparison*: refers to the tools with which the previous intervention is compared. In this case, there is no comparison as only MAS for the development of DT are investigated.
- *Outcomes*: refers to the overall outcome of the intervention. As we investigate how MAS are used to develop DTs, the findings identified are used as the outcome.

Using the above definitions, a search string was developed combining sections with "AND" statements as well as "OR" statements between each word within the section. "DT" was used as our search string jointly with two alternative spellings for MAS ("Multi-Agent" and "Multiagent"). Thus, the query will retrieve any paper related to MAS and DT. The final query was as follows: "DT" and ("Multi-agent" or "Multiagent").

Afterwards, according to Fig 2, the phase 2 was conducted. With this aim we used Science Direct, Scopus, IEEE, ACM and Springer as search engines. Only papers in which the search string appeared in the title, abstract or keywords were retrieved for the next phase. The query was initially conducted in June 2022, and run again in February 2023 to get the latest publications. It resulted in a corpus of 220 papers of literature related to DT and MAS. The papers found by the different search engines were then merged to remove duplicates.

Lastly, we applied the inclusion (IC) and exclusion criteria (EC). A paper will be selected to be reviewed if it meets all the inclusion criteria, while a paper will be excluded if it meets any of the exclusion criteria. The inclusion criteria were defined as follows:

- IC1: Papers that include multi-agent systems proposals to address the development of DTs.
- IC2: Papers in English.
- IC3: Peer reviewed papers in conferences and journals.
- IC4: Papers that provide information relevant to our scope.

On the other hand, the exclusion criteria defined are as follows:

- EC1: Papers in a language other than English (5 papers excluded).
- EC2: The study is a non-peer reviewed publication such as a technical report, editorial note, preface, index, introduction, presents a special / theme issue, or it is unreferenced, illegible, or inaccessible (e.g., paper is not available through the search engines used) (21 papers excluded).
- EC3: Books, unpublished papers, editorial notes, master's / degree works and doctoral theses as well as grey literature (blogs, web pages) (19 papers excluded).
- EC4: Papers duplicated (53 papers excluded).
- EC5: Papers that do not provide information relevant to our scope (50 papers excluded).
- EC6: Papers that are secondary studies (8 papers excluded).

Once the papers were checked by using the inclusion and exclusion criteria, a corpus of 64 papers, listed in Appendix A, was finally obtained. As indicated in [22], it is necessary to assess the quality of the studies included in a SLR to have confidence in the evidence provided and the conclusions drawn. Quality assessment (QA) criteria were established based on the following questions proposed in [22]:

- QA1. Are the objectives of the study clear?
- QA2. How adequately has the research process been documented?
- QA3. Does the study consider real-life applications?

When reading the full text of each study, points were awarded to each of the three evaluation criteria on a scale of 1 to 0. Each study is rated with a full point (1) when the information is concrete and reliable; it is rated with half-point (0.5) when the information is partially available; and it is rated with no point (0) when the information is not specified. For example, as described in [23], a full point (1) should be provided for QA1 if the objective of the study was described in the introduction (expected location); a half-point (0.5) should be given if the objective was stated vaguely or not at the expected location; and no point (0) should be provided if the objective of the study was not mentioned in the report. The mean score for each QA is: 0.91 for QA1, 0.91 for QA2, and 0.92 for QA3.

On other hand, to ensure the credibility of the selected studies, we set a minimum quality threshold of 1.5. Of the total corpus, 5 papers have a quality equal to 2, 22 papers have a quality equal to 2.5, and 37 papers have a quality equal to 3. Finally, the average quality of each paper is 2.75 out of 3. Therefore, any paper with a quality score lesser than or equal to 1.5 is ignored due to perceived lack of credibility.

5. Results

This section is structured in 5 Subsections, where the results of the analysis carried out to answer each one of the four research questions are presented.

5.1. Main statistics

Fig. 3(a) shows a breakdown of the corpus of papers according to their publication year. The large rise in the number of publications in the last years is remarkable. It is expected to continue to increase, since in 2023, considering only two months, there is already a meaningful number of papers compared with previous years. Fig. 3(b) shows a breakdown of the corpus of papers that have been included according to their type of publication. As can be observed, most of the included papers have been published in conferences.

Table 1 shows the publication channels with more than one published paper. The publication channel with the most publications is the conference *IFAC-PapersOnLine* whose publisher is *Elsevier*. As can be observed, there is not a specific channel for these papers.

Different application fields emerge from our analysis.

Table 2 shows the application domain for each paper included. The main application domain is manufacturing.

As mentioned in Section 1, one of the current challenges of DTs is how they can be designed and developed. Thus, identifying which tools and frameworks are being currently used to exploit the synergies of DT and MAS may help future developers of DTs. Once the corpus was analysed, we detected that, unfortunately, only 26.8% of papers comment on the framework(s) used for the development of their proposal. Table 3 summarizes the type of development technologies used in the analysed papers. Among these proposals, no standardized development technology was found. Moreover, none of the analysed proposals shows the modelling and design process of their DT.

5.2. RQ1. How do current MAS proposals support those properties expected in a DT?

By analysing the included papers, we have been able to identify the main properties of DT that are currently supported when used jointly with MAS. To perform this analysis we used the properties that Minerva et al. [2] gathered for DT, which were already described in Section 2. As aforementioned, these DT properties are classified into two groups: i) essential properties of DTs (DT1-DT3), and ii) properties that increase the value of DT included (DT4-DT12). In the following, the corpus of papers identified in Section 4 are analysed to determine how they support these DT properties.

Most of the researchers did not explicitly mention the properties of their developed DTs. However, most of them defined that is a DT that capture data from the PO to achieve a virtualization of them (LO) and that they need to collect those data that are relevant in their context of use. Also, they usually describe what objective they intend to

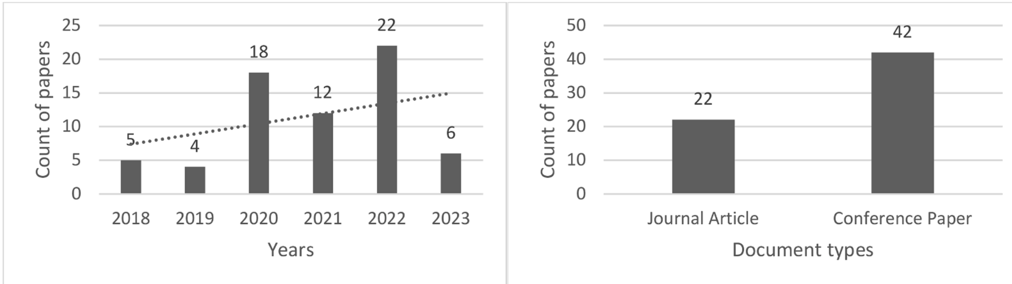

Fig 3. (a) Number of papers by year (b) Number of papers by type of publication.

Table 1
Number of inclusions per publication title.

Publication channel	Type	Publication source	Count
IFAC-PapersOnLine	Conference	Elsevier	4
Applied Sciences (Switzerland)	Journal	MDPI	3
IEEE Transactions on Industrial Informatics	Journal	IEEE	2
PervasiveHealth: Pervasive Computing Technologies for Healthcare	Conference	ACM	2
2018 Winter Simulation Conference	Conference	ACM	2
2022 2nd International Conference on Electrical Engineering and Mechatronics Technology, ICEEMT 2022	Conference	IEEE	2
Others	Conference and journals		49

Table 2
Summary of application domains for DT and MAS jointly applied.

Domains	References
Manufacturing: 25 papers (39.1%)	[8,13,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46]
Precision Agriculture: 8 papers (12.5%)	[15,47,48,49,50,51,52,53]
Energy: 7 papers (10.9%)	[54,55,56,57,58,59,60]
Smart City / Smart transport: 6 papers (9.4%)	[5,61,62,63,64,65]
Domain independent: 6 papers (9.4%)	[4,66,67,68,69,70]
Substations: 4 papers (6.3%)	[71,72,73,74]
Healthcare: 3 papers (4.7%)	[2,6,75]
Maintenance: 2 papers (3.1%)	[76,77]
Naval: 2 papers (3.1%)	[78,79]
Business and enterprise: 1 paper (1.5%)	[80]

accomplish with that data and how they use it. With this information it has been possible to identify the properties of each DT. For example, Croatti et al.[6] mentioned that "*the model represents all the features and aspects of the physical asset to which the DT is coupled through a cyber-physical connection properly defined.*". Thus, we concluded that the DT defined by them, fulfils the properties of *DT1-DT3*. Also, Croatti et al.[6] mentioned that “*Using this model, it is possible carrying out simulations about the current and future state of the asset, …, or continuously collecting and analysing data on the ongoing state of the asset to prevent unwanted situations.*”. Therefore, we concluded that the DT defined by them also fulfils the properties *DT6.Memorization* and *DT12.Predictability*. Appendix B shows a summary of the results of this analysis, identifying which properties are supported in each paper.

Not every DT property is supported by the analysed proposals. The three properties considered as essential (*DT1-DT3*) are supported by most of the analysed proposals, as was expected. Regarding the *DT1.*

Table 3
Summary of identified development technologies used for DT and MAS jointly applied.

Type	Framework / Platform	References
MAS Framework: 6 papers (9.4%)	JADE	[8,13,32]
	PADE	[34]
	JADEX	[28]
	JaCaMo	[6]
Own framework: 5 papers (7.8%)	Propose their own framework	[41,42,65,66,77]
Simulators: 3 papers (4.7%)	Simulator RepastHPC	[5]
	SDK	[63]
	MATLAB Simulink	[28]
Programming language: 3 papers (4.7%)	Java	[59,80]
	Python	[44]
Undisclosed framework: 47 papers (73.4%)	The framework used is not disclosed in the paper	[15,4,24,25,25,26,27,29,30,31,35,36,37,38,39,40,43,45,46,47,48,49,50,51,52,33,53,54,55,56,57,58,60,61,62,64,67,68,69,70,71,75,76,72,73,74,78,79]

RepresentativenessAndContextualization property when defining and creating a LO, it should be as similar as possible to its PO counterpart, representing in the LO all the attributes relevant according to the target context of use. Regarding the *DT2.Reflection* property, all significant attributes must be correctly and timely represented in the LO so that the state of the LO is the same as that of its PO. Finally, the *DT3.Entanglement* property is fundamental, since it refers to the communication between the PO and the LO, and the data sharing between them must be instantaneous. In short, in any design of a DT using a MAS, it must be as similar as possible to the real one, where its relevant attributes are represented in a timely manner and where the data exchange between them is in real time. For example, in the health field, Croatti et al. [6] proposal includes all these three properties. First, since the context of use of their DT is medical, and specifically trauma, they collect the patient’s medical data related to that trauma (*DT1. contextualization and representativeness*). Then, each of these relevant data of the person is timely reflected in the LO. These data are captured with sensors (*DT2. Reflection*). And finally, the person’s information collected by sensors is sent to the LO (*DT3.Entanglement*). Another example of DT that satisfies all these three properties is the one presented by Laryukhin et al. [49] in the field of modern precision farming. The authors develop software agents that work as DTs of real plants. Such agents constantly reflect the current phase of growth (*DT1. Contextualization and representativeness*) and development of the plant. Moreover, the relevant attributes of the plant are represented in the LO in a timely manner (*DT2.Reflection*). They also update their LOs daily with plant data captured by field sensors (*DT3.Entanglement*).

On the other hand, not all the properties of the second group defined to add value to the DT are supported in the papers analysed. The two properties of this group that are considered by most papers are *DT6. Memorization* and *DT12. Predictability*. The *DT6.Memorization* property describes the ability of DT to record all present and past data of the PO, so it is logical that many proposals consider it, because in DTs large volumes of data are generated for later use, for example, to make better decisions based on them. Croatti et al [6] support this property exploiting the DTs as a large source of information of a physical world. Another proposal that also considers this property is [33]. Their DT records past and present data of a production plant, facilitating data acquisition, data analysis and the use of such data to make informed decisions.

Next, *DT12.Predictability* property refers to the ability to simulate interaction with other POs or LOs and to predict the behaviour of the DT within an operational context of use. Again, it is normal that this property is addressed in many proposals because agents are undoubtedly characterized by their ease of interaction with each other to better accomplish their goals. Also, in [6], authors define agents that interact with their DTs, observing and acting on them. They also simulate possible interactions between different DTs, for example between DT of the patient and DT of the medical vehicle. Another example is [5], where the authors represent a smart city developed with DTs and MAS technology, simulating how people move throughout the city and how they interact with urban buildings and public transportation.

Some properties, such as *DT4.Replication, DT7.Composability* and *DT9.Augmentation,* are considered in just a few proposals. The first property, *DT4.Replication*, is included when the authors of the papers needed to create more than one LO for the same PO, because they were intended to be used in different contexts. For example, in [47] authors developed some plant LOs for precision farming management. Authors consider that the plant context changes according to its development stage, therefore, they create a specific LO for the same plant for each one of these stages.

The *DT7.Composability* property refers to the facility to group several DTs into a composite, so that the behaviour of the individual DTs and the composite one can be observed and controlled. In [53], in the field of greenhouse plant production, they define a DT of the greenhouse itself to capture the production process of the plants. With this aim, they compose the DT of the greenhouse with several subDTs, where each one represents a physical component of the greenhouse.

The *DT9.Augmentation* property is included in papers where the LO can be enriched with new attributes or functions needed in different contexts. For example, Ricci et al. [70] indicates that LO can provide other functionalities that conceptually augment the basic ones provided by the PO. Another example, the manufacturing industries are now moving towards mass personalization. In this sense, in [31], each personalized product is represented by a DT so that new attributes or functions can be added to the product to further customize it, increasing its functionality and adding value to the customer. In [69], authors design a DT as a personal assistant (called Personal DT, PDT) that extends the capabilities of the person and proposes a network of different PDTs that relate to each other as people do.

On the other hand, *DT5.Persistency, DT8.Accountability/Manageability*, and *DT10.Ownership* properties are considered each one in just one or two papers. As mentioned above, *Persistency* indicates that the DT should always be accessible over time. In [68] the author discusses a case study of DTs in the area of healthcare. These DTs operate when an anomaly is detected in the patient's clinical parameters, notifying the user to take the appropriate measures. For this reason, in this paper they highlight the importance of the DT always being available over time.

Regarding *DT10.Ownership*, there are two types of owners: on the one hand, the owner of the DT data, and on the other hand, the owner of the DT. In [69] authors refer to the second type by considering how personal DTs can substitute a person in some activities. The authors provide some facilities that can be used by a person to create his own LO, so that the owner of the DT is its creator.

Finally, *DT8.Accountability/Manageability* describes the ability to manage accurately and fully the DT. For example, the LO should provide self-management facilities for failure avoidance as well as for failure management facilitating its recovery by displaying all the most recent important functional values. For instance, in the manufacturing field, in [4] authors describe how real objects become intelligent objects by incorporating self-management capabilities. This facilitates that, if the DT perceives an anomaly in its behaviour, it can make decisions, for example, stopping the production to avoid further losses. And finally, DT11.Servitization is the only property not supported by any of the proposals analysed.

The paper that supports the most of the properties of DTs is [68], specifically eight properties. In that paper, authors introduce the concept of the Human DT (HDT), an agent whose goal is to align each person's goals with the intelligent machines around them. This is an interesting proposal given that more and more AI solutions are becoming ubiquitous in society. This paper considers both the three essential properties of DT (*DT1-DT3*) and the following properties that add value to DT: *DT4.Replication, DT5.Persistency, DT6.Memorization, DT9.Augmentation*, and *DT12.Predictability*. The suggested HDT endowed with all these properties allows to properly align AI with people's goals, achieving a better user experience in Human-Computer Interaction.

The other papers that meet more DT's properties are [6,15,47,48, 50–52,70]. These proposals include six properties of DTs. All these papers have modelled in their DT proposals the essential properties: *DT1-DT3*, as well as the properties adding value *DT6.Memorization* and *DT12.Predictability*. Croatti et al. [6] is framed in the field of health, the patients and their environment are represented in multiple LOs that can be observed and acted upon by agents. On the other hand, [15,47,48,50] is framed in precision farming, where each LO represents a growth stage of the plant. In these five papers the six properties named above are important, each LO represents a PO in a given context (*DT1.RepresentationAndContextualization*), relevant data are timely represented (*DT2.Reflection*), LOs and POs are connected to exchange information (*DT3.Entanglement*), all past and present data of the POs are stored (*DT6.Memorization*) and these data are used to simulate in the LO the future behaviour of the PO (*DT12.Predictability*). These last two properties belong to the set of value-adding properties of DT and are the most common ones in the analysed papers.

We have identified that most of the proposals that include property *DT6.Memorization* and *DT12.Predictability* aim to simulate the PO behaviour by using the collected data. However, not always the LO acts automatically on the PO. Thus, in these papers, a digital shadow is defined rather than an actual DT (Fig. 1). For example, Massel et al. [54] capture data from objects and energy systems to develop digital shadows. Authors claim that they plan add bidirectionality to their developments to be able to act on physical systems. Another paper that develop digital shadows is [15], they commented that plants' DT have limited functionalities focusing on monitoring the plant but they not are acting on the plant. However, in [51], the same authors already develop a DT (bidirectional connection, strong entanglement). The LO provides a continuous adaptive adjustment of plant development by being able to modify certain conditions such as lowering or raising the light level to which the plant (PO) is exposed.

Finally, from [24,25,29,30,39,42,54,60,62,65,72] it has not been possible to determine which properties of DTs support. This is because these papers focus their research on the design of an architecture for the integration of DT and MAS in industry, but they do not describe their own DT. On the other hand, Minerva et al. [2] discusses all the properties that a DT should have and suggest the use of multi-agent systems for its development, but do not describe how to actually use them to build DT.

5.3. RQ2. How are DT and MAS used in the development of applications?

The literature includes works that apply agents to model, design, implement or even exploit DT. This section analyses how the DTs and the MAS developed in the analysed proposals interact. Two main groups have been identified: *MAS with DT* and *MAS for DT*. Also, in this section we present the results of the analysis of how the properties of MAS and agents collected by Dorri et al. [3] can contribute to support the DT properties identified by Minerva et al. [2].

First, regarding how the DTs and the MAS developed in the proposals analysed interact. Two main groups have been identified:

- *MAS with DT*: From this point of view, DTs are considered as services that agents can exploit to sense and observe the state of POs of the environment, that is, MAS uses the DTs. In addition to the support for acting and observing, LOs also provide agents with facilities to control and manage the corresponding POs. In other words, from this perspective, the DT is a medium that provides agents with access to the physical world. An example, in the healthcare domain, is the LO that collects real-time patient information [6]. Then, the LO is observed and accessed by various agents acting as personal assistants for practitioners, such as doctors or nurses, involved in the treatment of the patient. Agents assist them in the treatment by showing updated patient's documentation and data.
- *MAS for DT*: From this point of view, the DT is developed using an adapted MAS architecture or using MAS as a component of the DT architecture, that is, MAS helps to develop the DT architecture. An example is [72], they proposed a five-dimensional DT model: (i) device layer, (ii) perception layer, (iii) network layer, (iv) platform layer and (v) an application layer developed as MAS. The objective of such application layer is to make decisions through predictive simulation and data analysis to improve the production of energy. Therefore, the agent is the decision maker and the learner in reinforcement learning. It works as follows: when it perceives the current state of the environment, it sends the next action instruction according to the learned strategy, that once it is completed, the environment issues a reward to the agent. Therefore, the smart behaviour of the DT is developed using a MAS.

Table 4 shows which papers belong to each group. As can be observed, most of the proposals focus on MAS that use DT for monitoring or managing the PO. Taking into account the type of interaction between MAS and DT, in the following subsections, we present the results of the analysis of how MAS and agent properties collected in [3], can contribute to support the DT properties identified in [2]. Moreover, based on the descriptions of the MAS properties and after analysing the papers included in this SLR, different design guidelines are also presented in the following to develop of DT using MAS and agents.

5.3.1. Agent1.Autonomy

As indicated in [81], this property is one of the most important properties of an agent because, as they claim, without autonomy an entity cannot be qualified as an agent. An autonomous agent can decide by himself what will be its next action, or whether it will perform it or not. The agent can make decisions internally according to its perceptions, knowledge, and beliefs. Therefore, it must be able, at least, to perceive and act to be qualified as autonomous.

DT8.Accountability/Manageability describes the ability to manage DT completely and accurately. Therefore, the agent' *autonomy* can help to model that property of DT. The principles of autonomous agent design can be a valuable contribution to create autonomous LO to make decisions about its *management/accountability* and ultimately carry them out without human intervention. For instance, in [4] they describe how

Table 4
Classification of the type of interaction between DT and MAS.

MAS with DT (43 papers)	MAS for DT (21 papers)
[73,43,52,51,74,70,71,57,58,78,79,77,45,46,27,66,4] [28,69,33,8,34,31,15,48,47,35,5,50,62,67,75,36,39] [63] [38] [37] [40] [30] [6] [55] [64] [61]	[65,41,53,72,76,60,80,44,59,42] [68,24,32,54,26,56,29,49,2,25,13]

PO become intelligent objects by incorporating self-management capabilities. This is a paper that belongs to the MAS with DT group. If the agent observes an anomaly in the DT data about the behaviour of the production chain, the LO can handle it, for example, by automatically stopping production to avoid further losses.

5.3.2. Agent2.Sociability

A social agent is an agent aware of other agents and capable of reasoning about them and interacting with them. According to Gleizes et al. [81], in a MAS, all agents have to be social, therefore, they must be able to interact with other agents and, perhaps, with people, sharing their knowledge and requesting information to improve their performance in achieving their objectives [3]. Thanks to this property, an agent supports providing and requesting information and services.

Recalling the properties of DT described in Section 2, the *sociability* of agents can help to model the *DT12.Predictability* property, because both refer to the interaction that can take place between agents and DTs. Therefore, classical agent interaction algorithms or methods can be applied to DT paradigm. There are many different proposals about social agents. In [6], authors define agents that interact with DTs, observing and acting on them (MAS with DT). Moreover, in the healthcare context, they also simulate the interaction between the DTs of the patient and the DTs of the medical vehicle, for example.

On other hand, *sociability* property contributes to *DT7.Composability* property that refers to the facility to group several objects into a composite. *Sociability* could help the various DTs to be grouped into one, facilitating interaction between the different LOs. Unfortunately, none of the papers identified in this SLR exploit this conjunction of properties.

5.3.3. Agent3.Reactivity

A reactive agent perceives and acts on its immediate environment and can respond in a timely manner to changes around it. In general, agents react to achieve their goals. Again, considering the *DT8. Accountability/Manageability* property refers to the ability to manage the DT accurately and completely. For example, among the proposals identified in Section 4, in [4] authors describe a DT that whenever it perceives an anomaly in its behaviour, it can stop production automatically to avoid further losses. This self-managed DT is a reactive agent that perceives and acts on its environment by responding in a timely manner to the changes occurring around it. This work is considered a MAS with DT as the agent observes the DT to detect those changes.

A reactive agent can detect a need that can be addressed by adding new attributes or functionalities. Therefore, this reactivity property may contribute to know when it is necessary the *DT9.Augmentation* property in the LOs. This property considers the use of data and new technologies, such as Big Data, to enhance the PO by adding new attributes and functions in the LO. Unfortunately, none of the papers identified in this SLR exploits this conjunction of properties.

5.3.4. Agent4.Proactivity

A proactive agent uses information from its history, detected parameters and information from other agents to predict possible future actions and make their own decisions to achieve their goals. This property implies that the same agent can perform disparate actions when they are in different environments.

DT12.Predictability property can be fostered by agent's proactivity because, among other things, refers to the ability of the DT to simulate the future behaviour of its PO in a specific environment. Therefore, by exploiting proactive agents it would be possible to predict how the PO would behave under specific conditions. In the field of smart agriculture, Skobelev et al.[15] claim that, among other things, with plant DT data, they create proactive agents whose goal is to predict plant productivity in a given situation. This work is a MAS with DT as the DT data helps to create the agent.

On the other hand, the DT8.*Accountability/manageability* property refers to the ability to manage the DT accurately and completely. In addition, the *DT5.Persistence* property refers to the DT being persistent and resilient over time to support a constant availability. In case of malfunction of the PO, the LO must be a source of information to restore and synchronize it to an acceptable state. A proactive agent must have access to its history, to its important functional values. Therefore, the way this data is stored and made available to the agent can be similar to the way a DT offers its data to its PO in case this needs it. Thus, proactivity can help to support the DT properties of *DT5.Persistence* as well as *DT8.Accountability/manageability*. For example, proactivity is also required in the previous example by Gorodetsky et al. [4]. When the LO detects an anomaly in the PO behaviour, and it stops production automatically to avoid further losses. Another example is presented in [68], where the DT acts when it detects an anomaly in the clinical parameters of the patient, alerting the user to take appropriate action. In both papers, the LOs designed with MAS are proactive as they respond to changes in their POs by deliberating and adopting goals to address the issues encountered. These two papers are included in the MAS with DT group.

Moreover, the *DT6.Memorization* property refers to the ability of the DT to record all the present and past data of the PO. A proactive agent uses information from its history to predict possible future actions, so the way this data is stored here can be a good approximation for storing the DT information. In [50], the LOs collect data in real time, that is, they are used as a knowledge base described using plant breeding ontologies. Then agents make decisions using such data to find the optimal solution in the process of planning rice growth stages. Thus, this is another example of MAS with DT.

5.3.5. MAS1.Leadership

The *leadership* property indicates that a MAS may have a lead agent, depending on the presence or absence of this type of agent, the MAS can be categorized either as leader or leaderless. In those MAS with no leader, each agent decides independently its actions according to its own objectives, while in those with a leader, this leader determines the goals for the rest of the agents.

The *DT4.Replication* property indicates that you can replicate a PO several times. According to Minerva et al. [2], over time, the set of attributes of each replica and the PO must be consistent. To synchronize them all, there should be a LO that acts as a master replica which is responsible for the synchronization of all relevant information. In cases where this master is needed to manage the different LOs created, the leadership principles studied in MAS can be very useful to provide the required artefacts to this master LO. In the papers identified in Section 4, no proposal was identified which exploits leadership to create such a master LO.

5.3.6. MAS2.Decision function

This property refers to how the input data affects the output of the agent's decision function. A MAS can be either linear or non-linear. In a linear MAS, the agent's decision is proportional to changes in the environment. Similarly, in a DT when data are captured from the PO sensors, their processing may produce a linear response or not. Therefore, defining this decision function for the DT inputs, alike it is done for agents, can enrich the design of the DT. For example, if an anomaly in the expected behaviour is detected, actions may be taken to avoid a major failure, these actions may be developed as decision functions, since detecting an anomalous value in the LO stops the operation of the PO. This self-management capability of the DT is referred in property DT8. *Accountability/manageability*. In [4], if the LO perceives an anomaly in the behaviour of the production chain (PO), using a decision function, it can automatically stop the production to avoid further losses.

On the other hand, *DT2.Reflection* determines that every significant attribute of the PO must be represented properly in the LO, so that the LO updates its state after collecting data. Depending on the input, the information captured can be reflected in the LO in a linear or non-linear way, depending on whether that information must be transformed before being stored. Unfortunately, we have not found any proposal that modifies the information before storing it.

5.3.7. MAS3. Heterogeneity

The *heterogeneity* property of the MAS refers to whether the agents that make it up have the same architecture, characteristics, and functionalities. If the architecture of the agents within the system differs, the MAS is heterogeneous. As indicated in [81], in a heterogeneous MAS, interoperability becomes one of the most important challenges to enable different agents to communicate to achieve their objectives. Different proposals have been presented to facilitate such interoperability. For instance, Wassermann et al. [82] provide a set of interoperability rules to improve the communication between heterogeneous agents. These proposals for effective communication can be good approximations to achieve correct communication between different DTs that make up a complete DT system. That is, solutions for heterogeneity can help to address the *DT7.Composability* property, improving the communication between the different individual DTs within the whole system. Although, there are no papers reviewed that include the *DT7.Composability* property in their DT proposals, it has been identified as one of the most desirable and challenging properties to be offered by DTs [83].

Also, a PO can be represented by different LOs replicas thanks to the *DT4.Replication* property. These LOs can have different attributes according to the *DT1.RepresentativenessAndContextualization* property. In the case that different LOs need to communicate with each other. Good communication practices between heterogeneous agents should be used to address these properties. Unfortunately, we have not found any proposal that replicate the PO in some LOs and them the LOs communicate with each other.

5.3.8. MAS4.Agreement parameters

This property refers to the number of metrics that the agents must agree upon to meet their objective. As indicated in [3], depending on the number of metrics, MAS are classified as first, second or higher order. In first order, agents collaborate to agree on a metric. In second order, agents must agree on two metrics, and a higher-order MAS, agreement between agents occurs when there are more than two metrics that converge to a common value.

The *DT4.Replication* property indicates that you can replicate a PO several times. Given that in different contexts different characteristics may be required to properly represent the PO of the DT, several replicas may be different as well. It is important for the different replicas to make consensual decisions when there is no a master or leader replica. Therefore, this property of MAS can be useful to facilitate this decision making. For example, in the same way as in a higher-order MAS, where the agreement between agents occurs when there are more than two metrics that converge to a common value, in the DTs replicas agreement can be reached between them when two or more make the same decision. Unfortunately, none of the papers included in this SLR have been identified as exploiting higher-order MAS to model DT.

5.3.9. MAS5.Delay consideration

This property enunciates that agents may face various obstacles that delay them in performing tasks. Depending on whether these delays are relevant or not, MAS can be classified into two groups: with delay or without delay. The former considers possible sources of delay, while the latter assumes that there are no sources of delay.

DT3.Entanglement property refers to the communication between LO and PO. The exchange of information between them must be instantaneous, however, in the real world, the exchange of information in a network generally introduces delays or even timeouts due to network failures that can appear. Therefore, studies that analyse the consensus and interaction between agents when there are communication delays can be useful to model *DT3.Entanglement* property. Also, the best practices and algorithms identified in MAS to reduce communication delays can be applied to the *DT3.Entanglement* property. For example, in [84], the authors study the synchronization problem of high-order linear agents sharing information in a communication network. According to this paper, the solution to this problem is to include a leader agent that imposes the desired behaviour to the rest of the agents and that can counteract the effects of communication deficiencies. Unfortunately, none of the papers identified in this SLR exploit this conjunction of properties.

5.3.10. MAS6.Topology

This property refers to the position and relations of the agents. MAS topology can be either *static*, where the position and relationships of an agent do not change during the agent's lifetime, or *dynamic*, where the position and relationships of an agent change as the agent moves, leaves, or joins the MAS, or establishes new relationships with other agents.

Again, considering DT properties, *DT7.Composability* refers to the ability to group several objects into a composite object and then observe and control the behaviour of the composite object as well as the individual components. Therefore, principles of dynamic topologies in which agents can be added to the MAS may be good options to approach the *DT7.Composability* property. One problem with the *DT7.Composability* property is how to organize the individual DTs in the composite, and a dynamic topology can help to determine which relationships the different LOs should keep. Unfortunately, none of the papers included in this SLR exploit dynamic topologies to model DTs.

5.3.11. MAS7.Data transmission frequency

The *data transmission frequency* property refers to how often information is shared in a MAS. There are two modes: i) agents collect data that share at predefined time intervals, and ii) agents collect data that share whenever specific events happen. The data transmission time delay often appears in communication networks. Undoubtedly, studies aiming at optimizing data transmission in MAS may be valuable contributions to develop the data exchange between LO and PO. For instance, Cui et al. [85] conclude that it is more optimal in MAS to share data just when certain events occur.

Fundamental to DT is the exchange of information between LO and PO, as described in the *DT3.Entanglement* property. Although Minerva et al. [2] mentions that the exchange should be in real time, depending on the intended use of the DT, it is not mandatory that the exchange of information is instantaneous, thus, it can be done whenever an event occurs or according to time intervals. For instance, Skobelev et al. [50] that propose sent data collected from the plants to the LO once a day or when certain events occur.

5.3.12. MAS8.Mobility

This property refers to the dynamism of the agents in a MAS. Agents can be classified as *static agents*, when they are always located in the same position in the environment, or *mobile agents*, when they can move in the environment.

Again, considering DT properties, *DT4.Replication* property describes the ability to represent a PO from the digital world with different LOs according to different target contexts of use. Mobile agents can move to different contexts to be more efficient or to use data, which for example, can be private and thus are not allowed to be accessed remotely. For example, in [15], each growth stage of a plant is represented in a LO that can change context. On the other hand, the *DT3.Entanglement* property refers to the exchange of information between LO and PO. The agent can be moved to facilitate communication between the PO (e.g. a user's sensors, as in [6]) and the LO that uses that information.

Also, with respect to *DT10.Ownership* property, it represents who is the owner of the DT data and both the physical and the digital objects. The agent may have to move to get data, and it is necessary to consider whether the data is owned or not to find out if it has access to this data. Unfortunately, none of the papers included in the SLR have been identified as exploiting the mobility to model DTs.

5.4. RQ3. What approach do current proposals use to manage knowledge?

One of the most important aspects for the DT is to know past and present data of the PO. The *DT6.Memorization* property defined by Minerva et al. [2] refers precisely to this. This property describes the ability of the LO to record those present and past data of the PO relevant for its behaviour. Considering *DT2.Reflection*, this data could be further transformed, or even combined from different sources, for its representation in the LO. This would also allow the LO making more informed decisions. Essentially, this data is used for two purposes: to understand the behaviour of the PO within its environment, and to predict its possible future behaviour. The property *DT6.Memorization* is fundamental to DT, since other properties base their functionality on it. For instance, *DT12.Predictability* needs data to make predictions, *DT8.Accountability/Management* and *DT5.Persistency* exploit data to facilitate data retrieval and autonomous management of the DT. Therefore, given the importance of managing data for a DT exploiting its full potential, it is necessary to investigate how currently DT and MAS proposals provide support for its management.

Table 5 shows the different approaches identified to manage knowledge in DT and MAS. First, it is worth mentioning that, as Table 5 shows, just 37.5% of the analysed papers comment about how they manage knowledge in their proposals. Moreover, we observed that, among such proposals, almost all of them (26.6%) use ontologies to manage knowledge. Ontologies formalise the structure of knowledge by defining concepts and relationships within a specific domain and they rely on an underlying database structure to store and retrieve information. On the other hand, DT is used as an element that facilitates the storage of knowledge, in other words, DT may act as extensive database, storing and facilitating access to data in real time. Although both ways of managing knowledge may be jointly used, no paper was identified in the corpus analysed that provides such facility. In the following, this aspect is further analysed.

5.4.1. Ontologies

As Table 5 illustrates, 26.6% of the papers use ontologies to manage knowledge. An ontology is a semantic component widely used in MAS that formally describes knowledge as a set of concepts within a domain and the relationships that hold between them. In the proposals analysed, ontologies are used to describe the behaviour, physical characteristics, etc. of the PO. Therefore, having this information embedded into the LO is essential, as it facilitates its decision-making process and supports simulating the future behaviour of the PO based on its past patterns (*DT12.Predictability*).

In [36], ontologies are used as a tool to manage the knowledge about the environment and the physical entities of the DT. To provide useful recommendations to humans, agents need access to a shared knowledge base that describes the properties of entities and the relationships between them. In the case of DTs, such knowledge base is a set of ontologies that specifies the underlying semantic models of the PO data. By integrating different ontologies, different types of related data can be linked through inference, and the ontologies that form the knowledge

Table 5
Summary RQ3: approaches to manage knowledge in DT and MAS identified.

Approach to manage knowledge	References
Ontologies: 17 papers (26.6%)	[4,13,15,29,30,33,36,44,47,48,49,50,51,52,54,59,80]
DT as knowledge base: 7 papers (10.9%)	[6,32,34,41,55,58,70]
Undisclosed: 40 papers (62.5%)	[68,69,24,25,26,66,27,28,8,31,2,56,64,35,61,5] [62,75,67,37,63,38,39,40,65,57,42,43,53,76,71] [78,72,73,60,74,79,77,45,46]

base can be extended and revised as new data arrives. In DTs, integrating ontologies from different POs contributes to the *DT7.Composability* property, since new attributes can be added to an ontology, and new ontologies can be added to the system by integrating new DTs.

On the other hand, ontologies also facilitate the interaction between LOs. In [29], the ontologies define the vocabulary and semantics of the knowledge used in the communication between LOs. This paper is an example of the *MAS for DT* group, since it integrates the ontologies (MAS itself) in their proposed architecture for DT. They propose a DT modelling method based on a multi-agent architecture, this architecture used as the base for DT models includes five components: (i) physical entities, (ii) virtual entities, (iii) MAS component, (iv) services and (v) semantic models. As indicated in [29], semantic models support the behavioural models of LOs. Also, ontologies enable agents to achieve their logical capabilities, such as reasoning and autonomous decision making, by formalizing the structure of knowledge of the entities, the relationships between them and the attributes of each one, etc.

Another paper using ontologies is [50] in the domain of modern agriculture, to support the decision-making process for agricultural production management. Authors propose the use of domain-specific ontologies for agricultural enterprises, crop production and precision farming tasks. For these authors, these ontologies should be at the foundation of the knowledge base of agricultural enterprises. Using this knowledge, they develop an ontology-based MAS that predicts plant growth, where different LOs represent different growth stage of the plant.

5.4.2. Digital Twin

Table 5 shows that 10.9% of the papers use DT to manage knowledge. These papers do not use MAS for the development of the DT, but they use the DT to provide facilities to the agents instead. In other words, the LOs of the different POs constitute large knowledge databases that agents can observe and thus make better decisions based on these data.

DTs and MAS of these papers belong to the *MAS with DT* group mentioned above (Section 5.3). For example, increasingly, customization is being pursued in product manufacturing. The MAS paradigm, on the one hand, looks promising for increasing the flexibility and adaptability of production systems. However, as indicated in [34], approaches to create these MAS often fail due to the huge effort required to create the knowledge bases of the individual agents and to ensure interoperability to communicate them. On the other hand, the DT concept provides the necessary means to create such knowledge bases, as they constitute a representation of the PO. Ocker et al. [34] study the combination of DTs with agents in the personalized production domain and conclude that a MAS is suitable for decision making in flexible production systems. The authors claim that the DTs of the respective physical entities provide an underlying KB for each agent, i.e., the DT information in a standardized form can serve as a knowledge base for an agent.

Another paper where the DT is seen as a large database is [6]. This paper also belongs to the *MAS with DT* group. In this case, any relevant asset has a digital counterpart (DT) modelled as part of the environment that agents can perceive and act upon. Thus, the DT collects real-time data provided by the asset's sensors and this data is used by the agents to make decisions based on this information.

6. Discussion

This section discusses the results of the review. We highlight observations from the results and investigate how this may affect future directions for the development of DT relying on MAS. For the sake of clarity, observations are summarized for each research question.

6.1. RQ1. How do current MAS proposals support those properties expected in a DT?

Most of the researchers did not explicitly mention the properties of their developed DTs. However, most defined that is a DT that capture data from the PO to achieve a virtualization of them (LO) and that they need to collect those data that are relevant in their context of use. Also, they usually describe what objective they intend to accomplish with that data and how they use it. With this information it has been possible to identify the properties of each DT. We have identified that most of the analysed proposals include all the essential DT properties: *DT1.Representativeness and contextualization, DT2.Reflection* and *DT3.Entanglement*. This seems logical, since in DT it is fundamental to adequately represent the relevant attributes of the PO in the LO and the connectivity between them is essential for information exchange. On other hand, from the set of DT properties that add value, the most common ones in the analysed proposals were *DT6.Memorization* and *DT12.Predictability*. This also makes sense, because data collected from the physical world are essential in LOs, for example to simulate the future behaviour of the PO.

The rest of the DT properties that add value are not included in most of the proposals because, depending on the context of use of the DT, it might not need to have all the properties. In other words, just as all the attributes of the digitized asset may not be necessary in the given context, not all the DT properties may be necessary. However, it is true that the more properties included, the more complete the DT will be. We believe that the lack of some properties such as *DT8.Accountability/Manageability* in the analysed proposals is justified by the fact that many proposals develop digital shadows without allowing the LO to operate on the PO.

6.2. RQ2. How are DT and MAS used in the development of applications?

Somers et al.[16] identified in their literature review that DT tends to favour passive testing techniques. Similarly, in this review it was noticed that DT just monitor the system during use. We have identified that generally the LOs of the current proposals obtain real-time data from the POs and then these data are used by the agents to make their decisions based on these data (MAS with DT). However, there is a need to move towards active and predictive behaviour to realize the full potential of this technology. Machine Learning (ML) is being applied in both DT [86] and MAS [87]. Undoubtedly, a future line of work is to apply ML models in DT and MAS developments to take advantage of the full power of both technologies.

Considering the analysis presented in Section 4, Table 6 was created to summarize which properties of DTs can be supported by using principles of agents and MAS properties. As can be observed, 91.7% of properties of DTs may be supported by MAS. The three properties that [2] consider essential for any DT are covered: *DT1-DT3*. On the other hand, only *DT11.Servitization* is not supported by MAS. Finally, as can be seen in Table 6, all the agent's and MAS' properties help to model some of the DT properties. This table also identifies whether any of the papers included has used an agent or MAS property to satisfy a DT property (☑) or not (☐). If there is a paper reviewed that includes this combination in their developments. Table 7 shows what agent and MAS properties can help to model each one properties of DTs and why these properties are related to each other. This table also identifies which references have exploited such MAS and agent properties. As can be observed, many of these properties are not exploited by any proposal what remains as an interesting future work.

We have identified that 91.7% of properties of DT are covered by MAS proposals. The three properties that Minerva et al.[2] consider essential for any DT are covered: *DT1-DT3*. Moreover, most of properties defined to add value to the DT are covered too. Only *DT11.Servitization* are not covered by any property of agents or MAS. On other hand, all the agent's and MAS' properties help to model some of the properties of DT. Thus, we confirm that MAS is a good approach to DT development.

Table 6
Properties of DTs that can be modelled with agent and MAS properties (☑ exploited ☐ not exploited).

	DT1	DT2	DT3	DT4	DT5	DT6	DT7	DT8	DT9	DT10	DT11	DT12
Agent 1								☑				
Agent 2							☐					☑
Agent 3								☑	☐			
Agent 4					☑	☑		☑				☑
MAS 1				☐								
MAS 2		☐						☑				
MAS 3	☐			☐			☐					
MAS 4				☐								
MAS 5			☐									
MAS 6							☐					
MAS 7			☑									
MAS 8			☑	☐						☐		

On the other hand, as can be seen in Table 7, in our analysis we have not found any paper reporting some combinations of agent properties and MAS to support DT properties. For example, we have not identified the exploitation of higher-order MAS for modelling DT in the reviewed papers. It would be important to take this into account when it is necessary to have several LOs for the same PO. If these replicas must agree to make consensual decisions, the principles of a higher-order MAS can help achieve such consensus. Therefore, given the synergies found between DT and MAS, designing a DT considering how its properties can be developed with agents and MAS properties looks a promising approach.

6.3. *RQ3. What approach do current proposals use to manage knowledge?*

Regarding the way knowledge is represented in the DT and MAS proposals, unfortunately only 37.5% of the papers indicate how they represent it in their implementation. We have identified that these proposals use one of these two approaches: i) ontologies, which are used as a database with information, properties, relationships about concepts that exist in the world or domain, and ii) DT, which are used as a database that contains updated information of the real asset.

We have identified that ontologies are the most common way to represent knowledge. There is a lack of MAS proposals that include another way of representing knowledge in the analysed papers. As Gayathri and Uma [88] state, the domain knowledge can be modelled in different ways to provide a complete description of the environment. Then, in addition to ontologies, authors name other techniques, such as Petri Net and Finite State Machine, that may be used to represent this knowledge. A line of future work is to analyse how these alternative forms could benefit the development of DTs and whether would help to achieve any of the properties of DTs.

On the other hand, we have identified in some papers that the DT represents the knowledge and agents have access to the data collected by the DT. However, we believe that a DT offers much more potential than simply being used as a database that agents can access. This stored data provides the *DT6.Memorization* and in addition, this data can be used by machine learning algorithms to support other properties such as *DT12.Predictability* [86]. If a DT is designed with the objective of predicting the behaviour of the PO using the LO, these two properties of the DT (*DT6.Memorization* and *DT12.Predictability*) should certainly be included. We believe that these two value-adding properties should be in all DTs. Unfortunately, some of the included papers do not include them.

On other hand, Kulmanov et al. [89] commented that ontologies are increasingly included in machine learning models. Including ontologies that provide more information about the domain can also be very interesting to optimize machine learning algorithms. Although ontologies are the most common representation alternative in the analysed papers, none of them use ontologies for machine learning models, therefore, we have identified here a potential niche for future work.

6.4. *Research agenda*

In the following a research agenda about the synergies between MAS and DT is presented, describing first in Section 6.4.1 a summary of the research conducted during the last twenty years. Then, Section 6.4.2 presents the current landscape of MAS and DT, considering specially the DT properties. Section 6.4.3 identifies which are the most recent trends in the area. Lastly, in Section 6.4.4, a summary of the most streamline future research topic is presented.

6.4.1. *Reflection on the research of the past 20 years*

Since the origin of the DT in 2003, academic interest has been growing over the last 20 years, with exponential growth in recent years. The number of published papers related to DT and MAS has also been growing in parallel. The integration of DT and MAS represents a growing area of research aimed at developing advanced applications and improving the DT functionality in various domains. The papers included in the corpus analysed mainly include practical cases of domain-specific DT, without providing a theoretical foundation for the development of DT with MAS. Our analysis of the corpus indicates that this is a reflection of the pervasive relevance of the manufacturing and industrial domain. Few theoretical papers have been found that provide a basis on how to develop DT supported by MAS. Therefore, it can be claimed that the area of *MAS for DT* can become mature as a research discipline only if efforts are made to provide a solid theoretical basis. Moreover, we identified that the analysed research on DT and MAS does not follow a domain-agnostic proposal that guides the design of DT. In this regard, there is also a need for the research community to join efforts to provide methodologies that facilitate the design of DT using MAS as the development approach. The release of new frameworks customized for guiding and scaffolding both DT design and development using MAS methodologies would also be a step towards the maturity of the DT field.

6.4.2. *Current structure of the research landscape*

Current research on DT development using MAS considers less than 50% of the properties of DT in its developments (see Appendix B). Most of the analysed proposals include the three essential DT properties: *DT1. Representativeness and contextualization, DT2.Reflection* and *DT3.Entanglement*; and the add-value properties *DT6.Memorization* and *DT12.Predictability*. Our analysis of the literature reveals that the lack of some properties, such as *DT8.Accountability/Manageability,* in the analysed proposals is explained by the fact that many proposals develop digital shadows without allowing the LO to operate on the PO rather than true digital twins.

On other hand, most of the articles included in the analysed corpus mainly include case studies of domain-specific DT and most of them focus on *MAS with DT*, that is, the DT is used to monitor or manage the physical part of the DT (PO). This trend reveals that current research has a bias towards the development of digital shadows rather than DTs. Digital shadow refers to a digital representation that captures and reflects real-time data from the PO, but without the bidirectionality or

Table 7
Summary of the synergies between agent, MAS, and DT properties.

DT's properties	Agent and MAS' properties	Synergies between both	Related papers
DT1. Representative and contextualization	MAS 3. Heterogeneity	LOs can have different attributes depending on the context of use, giving rise to a heterogeneous system.	None
DT2.Reflection	MAS 2. Decision Function	Depending on the input, the information captured of PO can be reflected in the LO in a linear or non-linear way, depending on whether that information must be transformed before being stored.	None
DT3.Entanglement	MAS 5.Delay Consideration	The exchange of information between LO and PO should be instantaneous, however, in the real world, the exchange of information in a network usually introduces delays or even timeouts due to network failures.	None
	MAS 7.Data Transmission Frequency	The exchange of information between LO and PO can be done in real time, when an event occurs or according to time intervals.	[50]
	MAS 8.Mobility	The agent can be moved to facilitate communication between the PO and the LOs.	[6]
DT4.Replication	MAS 1. Leadership	Since a PO can be represented by several LOs, a master LO (leader) should be designated who is responsible for the synchronization of all of them.	None
	MAS 3. Heterogeneity	A PO can be represented by different replicas of LOs giving rise to a heterogeneous system	None
	MAS 4. Agreement parameters	It is important that the different replicas make consensual decisions when there is no master or leader replica, therefore, this property of MAS can be useful to facilitate this decision making	None
	MAS 8.Mobility	Mobile agents can move to different contexts to be more efficient or to use data, which for example may be deprived of an environment.	None
DT5.Persistency	Agent 4. Proactivity	A proactive agent can help the DT to restore its data if needed in case of failure.	[15]
DT6.Memorization	Agent 4. Proactivity	A proactive agent uses information from its history to predict possible future actions, so the way this data is stored here can be a good approximation for storing the DT information	[50]
DT7.Composability	Agent 2. Sociability	Sociability could help the different DTs to be grouped into one, facilitating the interaction between the different LOs.	None
	MAS 3. Heterogeneity	Can help to improve communication between the different individual DTs within the whole system	None
	MAS 6. Topology	A dynamic topology in which new agents can be added to the MAS can be a good option for organizing the individual DTs in the composite.	None
DT8. Accountability / Manageability	Agent 1. Autonomy	DT must include functionalities for its self-management, therefore, autonomy property of MAS can contribute to address such DT requirement.	[4]
	Agent 3. Reactivity	A self-managed DT can be a reactive agent that perceives and acts on its environment by responding in a timely manner to changes around it.	[4]
	Agent 4. Proactivity	A proactive agent can help the DT to restore its data if needed in case of failure.	[68]
	MAS 2. Decision Function	In a DT, data is captured from the PO through sensors, and these data are analysed and may or may not produce a linear response when processed	[4]
DT9.Augmentation	Agent 3. Reactivity	A reactive agent can detect a need and by adding new attributes or functionalities a response can be provided. The LO can provide other functionalities that augment those provided by the PO.	None
DT10.Ownership	MAS 8. Mobility	When the agent moves to get data, it is necessary to consider the ownership of the data. Privacy issues are relevant in many application domains.	None
DT12.Predictability	Agent 2. Sociability	Sociability of agents can facilitate the interaction between DTs.	[6]
	Agent 4. Proactivity	Using proactive agents, it would be possible to predict how the DT would behave under specific conditions.	[15]

depth of interaction that characterizes true DT, which is designed to be complete and dynamic software representations capable of simulating, predicting, and optimizing the behaviour of the PO through two-way interaction. Here we can observe an apparent lack of studies that develop true DTs, where the physical world can be operated on from the virtual one.

This trend suggests an important area of opportunity for future research and development in the field of DT and MAS. There is a potentially gap to explore in the design and implementation of DTs that not only reflect the state of the PO, but also interact with it in a proactive way to improve its performance and efficiency. Furthermore, the current literature could benefit from further exploration of how the add-value properties of DT can be harnessed in different domains, thus extending the reach and impact of these systems in industry and society.

Finally, regarding DT properties supported by using the properties of agents and MAS, we found some interesting observations. Thus, we believe that the development of *MAS for DT* will continue to be a relevant topic both in practice and in academia for the next years. Table 7 shows which agent and MAS properties can help to model each one of the DT properties and why these properties are related among them. This table also identifies which references have exploited such MAS and agent properties. As can be observed, many of these properties are not exploited by any of the analysed proposal, identifying interesting niches for future work.

6.4.3. New emerging research themes

Early research on DT and MAS started with related contributions in the field of industry and consisted mainly of digital shadowing. Fortunately, the most recent literature has expanded the field providing new advances and identifying novel areas of research. Firstly, one promising area is the integration of artificial intelligence (AI) with DT and MAS systems to significantly improve the autonomy and decision-making capabilities of the systems. AI can empower DTs to not only simulate and model behaviours, but also to learn from past interactions and predict future needs and potential failures in real time.

On other hand, DT and MAS research is opening new opportunities up that could transform many sectors. Among them, the exploration of autonomous self-repair systems that combine DT and MAS to maintain and repair critical infrastructures without human intervention emerges as an attractive research field. Also, the synergy of DT and MAS could help in the design and development of a Human Digital Twin (HDT) for its exploitation in different domains such as healthcare, entertainment, marketing, etc. Another emerging theme is the development of DT and MAS for global logistics management, optimizing supply chains in real time. In addition, the use of these technologies for smart city design, where DT and MAS work together to manage everything from traffic to utilities and security, represents a field of great potential.

6.4.4. Summary of future challenges

After analysing 64 primary studies related to DT and MAS, we identified some open challenges for future research on these technologies. As a contribution, we list seven topics to define a research agenda. These topics were defined according to the conclusions and gaps identified from the results of this secondary study:

- Definition of novel theoretical proposals that provide a basis on how to develop DT supported by MAS.
- Development of frameworks that allow a guided design of DT using MAS methodologies.
- Investigating how the properties of DT can be supported by the properties of agent and MAS to improve the DT.
- Development of DT rather than just digital shadows, that is, development of DT that support both the essential and the add-value DT properties.
- Formulating DT proposals that support the value-adding DT properties by exploiting the properties of agents and MAS.
- Development of DT and HDT using MAS in different domains such as smart cities or e-health.
- Design of ontologies for machine learning models that can be used by DT.

7. Threats to validity

Following the guidelines for conducting SLR presented in [22], in the following the threats to validity that may have affected the results presented in this SLR are analysed. The categories of threats to validity defined by Zhou et al. [90] have been analysed:

- *Construct validity* refers mainly to the review protocol and the degree of inclusion of the selected search string, data sources, and selection criteria with respect to the research questions. To minimize the threats to construct validity, a validation procedure was included for each step of the review protocol. For instance, the search string was defined as generic as possible not to exclude any potentially interesting studies following PICO guidelines. However, it is possible that the search did not include some relevant studies due to the choice of search terms, data sources and selection criteria.
- *Internal validity* refers to the reliability of the cause-effect relationship between the results and the methodology of the review. In this SLR, the search procedure was mainly performed by a single researcher, so the results may be subject to selection bias. To reduce this, a validation procedure was performed in which the other authors reviewed a randomly selected set of papers, so that each paper was reviewed by both the first author and at least one other author. In addition, a paper was only rejected if both authors were sure that it was irrelevant according to the inclusion and exclusion criteria. In case of doubt, the paper passed to the next stage and, if the review of the full text the decision did not offer a clear result, the other authors reviewed it independently and discussed it until a group consensus was reached.
- *External validity* refers to how the results of the review are applicable to the subject area, which in this case is DT development using MAS. Due to the lack of a comprehensive definition of DTs, as stated in Section 2, we have followed the definition of DTs provided by Minerva et al. [2]. It could be argued that this review may not be valid for those DTs that do not fit the definition used in this review. To minimize this threat to validity, in Section 2 we have detailed the definition of DT used in this SLR to properly describe the scope in which this review is valid.
- *Conclusion validity* refers to the reproducibility of the results. In order to maximize the reproducibility of our results, the guidelines for SLR defined in [22] were used to provide all the necessary details including choice of search string, data sources, and selection criteria. However, some subjectivity may still exist while extracting information for each research question so that, in case of replication, it cannot be ensured that the same conclusions may be drawn.

8. Conclusion and future work

In this paper we present the results of a systematic literature review on DT and MAS aimed at understanding how MAS can support DT development. It is undeniable that DT and MAS are two terms that have attracted the interest of the research community in recent years. We have provided an overview of the current MAS and DT landscape, as well as outlined suggestions and future research areas in this field. The systematic process followed led us to identify a collection of 220 papers related to DT and MAS in the search engines Science Direct, Scopus, IEEE, ACM, and Springer. Once these papers were filtered a final corpus of 64 papers was selected. Using these papers, our study has answered five research questions.

First, with respect to RQ1, we have identified that most of the current MAS and DT proposals support the essential properties of DT, and for the remaining properties that add value, two of them stand out as being the most supported ones in the MAS for DTs analysed: *DT6.Memorization* and *DT12.Predictability*. On other hand, most of the DT proposals do not define how DT operates and modifies the physical world, therefore, they are digital shadows rather than full DTs.

Second, regarding RQ2, we have identified two approaches according to the type of interaction between DT and MAS: MAS with DT and MAS for DT. Moreover, we have identified that every MAS property has been exploited for supporting at least one DT property. Only the property *DT11.Servitization* is not satisfied by any MAS property. This is justified because this property is related to business but not to technology.

Third, with respect to RQ3, we have identified that the most common way of representing knowledge in the analysed proposals is ontologies. They use ontologies to manage the knowledge of the physical world in the cyberworld, that is powered by the real data obtained by the DT. In such cases, ontologies are used as a tool to digitalize the knowledge of the environment and of the physical objects needed by the agents to make decisions. In addition, this way of using ontologies to manage knowledge is used to improve and facilitate the interaction between agents. Moreover, DTs are sometimes seen as a database. In these cases, the DTs of the different physical entities represent big knowledge databases that agents can observe, make decisions based on this information and act on the DT as well.

On other hand, based on our systematic literature review, we have identified some opportunities for further research and work, considering different research gaps. In the following the most relevant ones are presented:

First, most of the reviewed papers include less than 50% of the properties of DT in their developments. Therefore, one avenue for future work is to develop more complete DTs by addressing more DT properties using agent and MAS properties.

Second, all the agent and MAS properties have been used by the included papers to model some of the DT properties (see Table 6). Therefore, conducting a formal evaluation of how such MAS properties may be exploited to support the DT properties would be also interesting future research work.

Third, as shown in Table 6, we have identified that some agent and MAS properties have a high potential to satisfy some of the most challenging DT properties. Thus, an interesting future work would be to analyse the use of such agents and MAS properties for the development of DTs.

Fourth, we have identified that in the development of DT and MAS, ontologies are the most used way to represent knowledge. In addition, as was described in Section 6.3, the exploitation of AI is key for providing DTs with its full potential. Therefore, a future research gap would be the use of ontologies along with machine learning algorithms to provide more information to the models. No paper has been identified that follows this approach.

CRediT authorship contribution statement

Elena Pretel: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Alejandro Moya:** Validation, Investigation. **Elena Navarro:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Víctor López-Jaquero:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. **Pascual González:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

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